Customer Shopping Behavior Analysis

A comprehensive end-to-end data project integrating **Python** for data cleaning, **SQL** for structured querying, and **Power BI** for interactive visualization — transforming raw customer data into actionable business insights.

DATA ANALYTICS

RETAIL INTELLIGENCE

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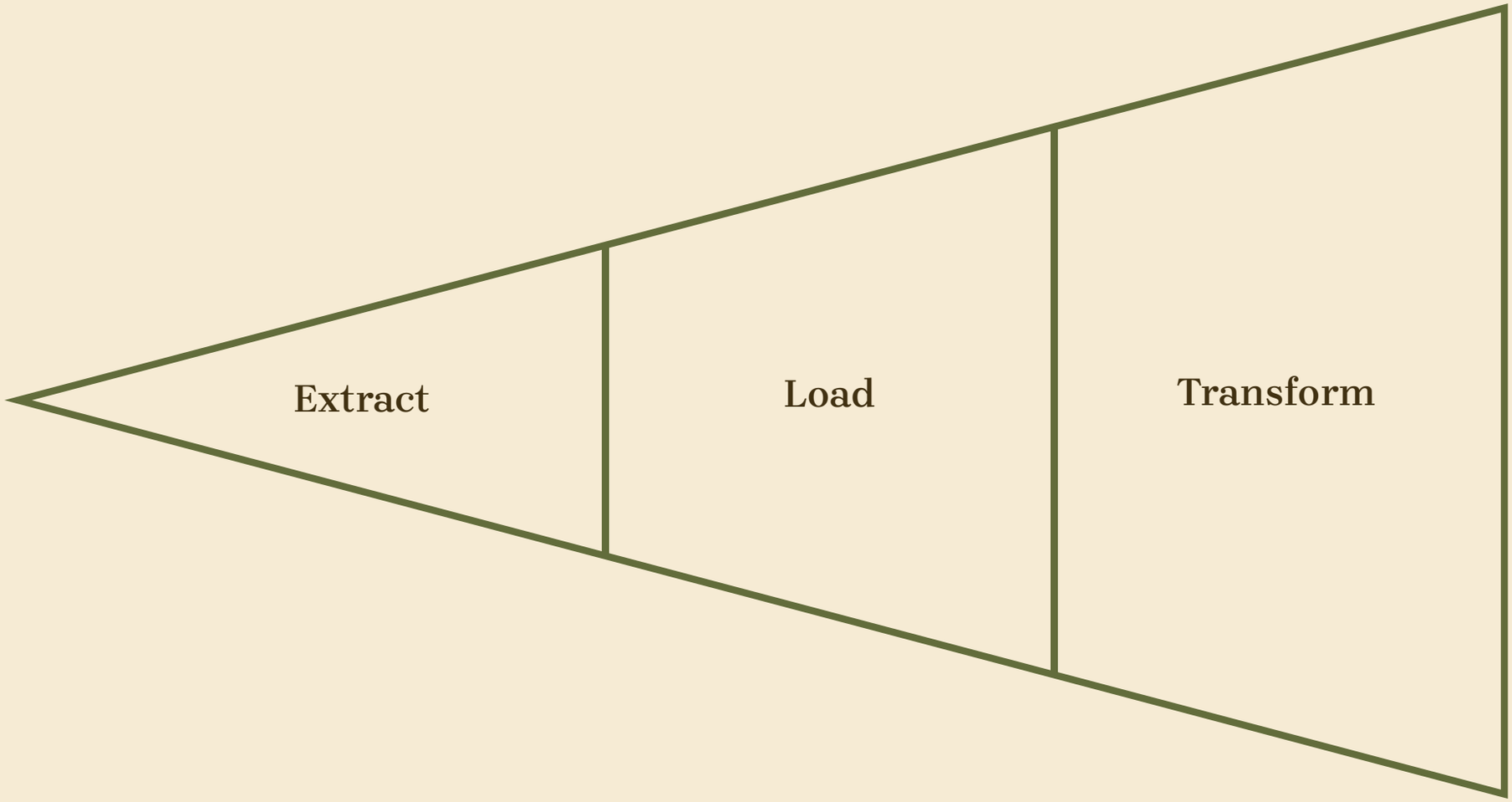
Project Overview

The objective was to analyze customer purchase patterns, demographics, and preferences to derive actionable business insights. The project followed a structured, four-stage data pipeline.



Each phase built upon the last — ensuring clean, structured, and visually accessible insights for stakeholders.

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Phase 1: Data Preparation & Cleaning

Using a Jupyter Notebook (`Customer_Shop_behaviour.ipynb`), the raw dataset was processed to ensure data integrity and create new analytical features.

Data Inspection

Checked data types, missing values, and statistical distributions using `df.info()` and `df.describe()`.

Missing Values

Imputed missing `review_rating` values with the median rating within each product category.

Standardization

Converted column names to lowercase with underscores; renamed `purchase_amount_(usd)` to `purchase_amount`.

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Feature Engineering & Redundancy Removal

Age Grouping

Customers were binned into four meaningful segments to support demographic analysis:

- Young Adult
- Adult
- Middle-Age
- Senior

Frequency Mapping

Categorical purchase frequencies (e.g., "Weekly") were converted into numeric days to support time-based analysis.

Redundancy Removal

The `promo_code_used` column was dropped after verifying it was perfectly correlated with `discount_applied`, eliminating redundant data.

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Phase 2: Database Management & SQL Analysis

The cleaned data was exported to a MySQL database (`Customer_Behaviour`) to perform advanced analytical queries addressing key business questions.

01

Revenue by Gender

Male customers contributed significantly higher total revenue than Female customers in this dataset.

02

Customer Segmentation

Customers categorized as **New** (1 purchase), **Returning** (2–10), or **Loyal** (11+ purchases).

03

Subscription Impact

Compared average spend and total revenue between subscribers and non-subscribers.

04

Top Products

Identified top 3 most purchased items per category using window functions (`ROW_NUMBER()`).

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Phase 3: Power BI Dashboard

An interactive dashboard (customer_behavior_dashboard.pbix) was created to present findings to stakeholders across three core focus areas.

Sales Performance

Tracking revenue trends across seasons and product categories.

Customer Demographics

Analyzing contribution of different age groups and genders to total revenue.

Loyalty & Retention

Visualizing the distribution of New vs. Loyal customers across the base.

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Visual Insights: Revenue & Demographics

Revenue by Gender

Analysis shows the distribution of total spending across genders, helping target marketing efforts more effectively.

Revenue by Age Group

Identifying which age demographic spends the most allows for better product curation and inventory planning.

Gender Split

Male customers drove higher total spend in this dataset.

Age Segments

Young Adults emerged as the primary revenue-driving demographic.

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Visual Insights: Loyalty & Top Products

Customer Segmentation Distribution

This chart segments the user base by loyalty tier. A large portion of **"Loyal"** customers indicates high retention potential and strong brand stickiness.

Top Rated Products

A look at the highest-rated items according to customer feedback, informing future inventory and promotional decisions.

New

1 purchase

Returning

2–10 purchases

Loyal

11+ purchases

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Key Findings

1

Loyalty Matters

The majority of the customer base falls into the **"Loyal"** segment, suggesting strong brand stickiness and high retention potential.

2

Young Adults Drive Revenue

The **"Young Adult"** group is a primary revenue driver, suggesting inventory should be curated to their preferences.

3

Subscription Opportunity

Non-subscribers showed **higher average spend**, indicating an opportunity to convert "big spenders" into subscribers through targeted incentives.

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Conclusion & Takeaways

This project demonstrated a complete, production-ready data analytics pipeline — from raw CSV ingestion to an interactive Power BI dashboard — delivering measurable business value.

Python

Clean, engineered data ready for analysis

SQL

Structured queries revealing revenue and loyalty patterns

Power BI

Interactive dashboards enabling stakeholder decisions

The integration of Python, SQL, and Power BI provides a scalable, repeatable framework for ongoing customer behavior analysis and strategic retail decision-making.

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